

three generations of matter (fermions)

interactions / force carriers (bosons)

I

II

III

mass

$\approx 2.16 \text{ MeV}/c^2$

$\approx 1.273 \text{ GeV}/c^2$

$\approx 172.57 \text{ GeV}/c^2$

0

$\approx 125.2 \text{ GeV}/c^2$

charge

$\frac{2}{3}$

$\frac{2}{3}$

$\frac{2}{3}$

0

0

spin

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

0

u

c

t

g

H

up

charm

top

gluon

higgs

d

s

b

γ

down

strange

bottom

photon

e

μ

τ

Z

electron

muon

tau

Z boson

ν_e

ν_μ

ν_τ

W

electron
neutrino

muon
neutrino

tau
neutrino

W boson

QUARKS

LEPTONS

GAUGE BOSONS
VECTOR BOSONS

SCALAR BOSONS